How-To Library

Informatica

Implementing Informatica® Big Data Management in a Microsoft Azure Cloud Environment

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Abstract

You can take advantage of cloud computing efficiencies and power by deploying the Informatica Big Data Management solution in the Microsoft Azure environment. You can use a hybrid solution to offload or extend on-premises applications to the cloud. You can also use a lift-and-shift strategy to move an existing on-premises big data solution to the Azure HDInsight environment to improve processing speed. This article describes how to implement the one-click Big Data Management deployment in the Azure environment.

Supported Versions

- Informatica Big Data Management 10.1.1 Update 2

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Overview

Customers of Microsoft Azure and Informatica can deploy Big Data Management in the Azure public cloud.

Use the Azure Marketplace to install a new instance of Big Data Management in the same virtual network or vnet where you run an HDInsight cluster.

When you install Big Data Management from the Azure Marketplace, you use your existing HDInsight cluster. You choose the size of the machine for the Informatica domain database.

After you complete installation, use Big Data Management to run mappings and workflows on the HDInsight cluster.

Using Big Data Management on Azure provides the following benefits:

- **Faster time to insight.** Dynamic big data integration delivers high throughput data ingestion and data delivery from nearly any source, leveraging Azure for high performance data processing at scale, and delivering the right analytical data to business stakeholders.
- **Faster time to deployment.** The simple One-Click automated deployment of Big Data Management on the Azure Marketplace allows organizations to quickly and efficiently deploy a big data integration solution on a high-performance cloud infrastructure platform.
- **Accelerated data architecture modernization.** If you are planning to modernize your data strategy initiatives on Azure, Big Data Management provides rich functionality, such as metadata driven data integration, dynamic mappings, and SQL to mapping conversion to help shorten development cycles and reduce time to market.
- **Clean, complete, and trusted data.** Whether you are offloading or extending on-premises applications to the cloud or fully embracing the cloud, collaborative data quality ensures confidence in data fidelity while facilitating data sharing, empowering business stakeholders to curate data, audit data holistically, and relate data at scale. Big Data Management empowers organizations with complete, high-quality, actionable data.

Pre-Implementation Tasks

Before you provision Azure resources and configure Big Data Management in the Microsoft Azure cloud environment, verify the following prerequisites:

Prerequisites

- You have purchased a license for Big Data Management.
  The license file has a name like `BDMLicense.key`. During configuration, you browse to this file on your local system. It is not necessary to upload it to the Azure environment.
- You have an Azure subscription and its account authentication information.
- You have created an HDInsight WASB cluster in the Azure environment.

Note: When you provision Big Data Management, you must use the virtual network or vnet where the current instance of HDInsight is configured.

Gather Cluster Information

Gather the following information from the cluster, and input the information during configuration of the Big Data Management instance:

- Ambari host machine, port number, and user authentication details
- SSH port and user authentication details

Create Staging Directories on the Cluster

In the HDFS storage location, create the following directories on the cluster and set permissions to 777:

- - Blazeworkingdir
  - SPARK_HDFS_STAGING_DIR
  - SPARK_EVENTLOG_DIR

Provision Azure Resources and the Informatica Domain

Use the Azure Marketplace website to provision Azure cluster resources including a Big Data Management deployment.

1. Select Big Data Management for setup.
   a. In the Azure marketplace, click the + button to create a new resource.
   b. Search on "Informatica" to find Informatica offerings in the Azure marketplace.
   c. Select **Big Data Management 10.1.1 U2 BYOL.**

      The Create Big Data Management Enterprise Edition tab opens. It displays all the steps necessary to configure and launch Big Data Management on an HDInsight cluster.
2. Supply information in the **Basics** panel, and then click **OK**.

   Subscription

   Select the Azure subscription account that you want to use for Big Data Management.

   Charges for this instance of Big Data Management will go to this subscription.

   Resource Group

   Select a resource group to contain the BDM implementation.

   Usually, you select an existing resource group where you have a running HDInsight cluster.

Location

Location of the resource group.

Accept the location that is already associated with the resource group.

3. Supply information in the **Informatica Domain Settings** panel, and then click **OK**.

   This tab allows you to configure additional details of the Informatica domain. All properties in this tab are mandatory.

   Informatica Domain Name

   Create a name for the Informatica domain.

   Informatica domain administrator name

   Login to use to administer the Informatica domain.

   Password

   Password for the Informatica administrator.

   Keyphrase for encryption key

   Create a keyphrase to create an encryption key.

   Informatica license file

   Click the Folder icon to browse to the location of the Informatica license file on your local system.

   When you select the license file and click OK, Azure uploads the file.

4. Supply information in the **Node Settings** panel, and then click **OK**.

   This tab allows you to configure details of the Informatica domain.

   Select the OS for the VM.

   Select **Red Hat Enterprise Linux 7.3**.

   Number of nodes in the domain.

   Default is 1.

   You can configure up to 29 nodes.

   Machine prefix

   Type an alphanumeric string that will be a prefix on the name of each virtual machine in the Informatica domain.

   For example, if you use the prefix "infa" then Azure will identify virtual machines in the domain with this string at the beginning of the name.

   VM Username

   Username that you use to log in to the virtual machine that hosts the Informatica domain.

   Authentication type

   Authentication protocol you use to communicate with the Informatica domain.

   Default is **Password.**

   Password

   Password to use to log in to the virtual machine that hosts the Informatica domain.

   Machine size

   Select from among the available preconfigured VMs. The default is 1x Standard DS11.

5. Supply information in the **Database Settings** panel, and then click **OK**.

This tab allows you to configure settings for the storage where Informatica metadata will be stored.

Database type

Select **SQL Server 2014.**

Database machine name

Name for the virtual machine that hosts the domain database.

Username

Username for the administrator of the virtual machine host of the database.

These credentials to log into the virtual machine where the database is hosted.

Password

Password for the database machine administrator.

Database machine size

Select a size from among the available preconfigured virtual machines. The default is 1x Standard DS3.

Informatica Domain DB User

Name of the database user.

The Informatica domain uses this account to communicate with the domain database.

Informatica Domain DB Password

Password for the database user.

6. Supply information in the **Informatica Big Data Management Configuration** panel, and then click OK.

This tab allows you to configure credentials that allow the Informatica domain to communicate with the HDInsight cluster. Get the information for these settings from HDInsight cluster settings panels and the Ambari cluster management tool.

HDInsight Cluster Hostname

Name of the HDInsight cluster where you want to create the Informatica domain.

HDInsight Cluster Login Username

User login for the cluster. This is usually the same login you use to log in to the Ambari cluster management tool.

Password

Password for the HDInsight cluster user.

HDInsight Cluster SSH Hostname

SSH name for the cluster.

HDInsight Cluster SSH Username

Account name you use to log in to the cluster head node.

Password

Password to access the cluster SSH host.

Ambari port

Port to access the Ambari cluster management web page. Default is 443.

7. Supply information in the **BDM Services Settings** panel, and then click **OK**.

Use this tab to set up Informatica services. For more information about Informatica services, see the *Informatica 10.1.x Application Service Guide*.

Model Repository Service Name (MRS)

Name of the Model Repository Service.

The Model Repository Service manages metadata about mappings, workflows and applications.

Data Integration Service Name (DIS)

Name of the Data Integration Service.

The Data Integration Service runs mappings and workflows.

MRS Database username

Username for the Model Repository Service database.

The Model Repository Service database stores Model Repository data.

MRS Database password

Password for the Model Repository Service database.

8. Supply information in the **BDM Connection Settings** panel, and then click **OK**.

Use this tab to configure connections between the Informatica domain and the HDInsight cluster. The Data Integration Service uses the connections to run mappings on the cluster.
The HDFS User Name property is mandatory. The other properties in this tab are optional.

For more information about connections, see the Connections appendix of the *Informatica Big Data Management User Guide*.

HDFS User Name

User name for the HDFS connection.

HBase Zookeeper Hosts

List HBase Zookeeper hosts, separated by the comma (,) character.

Hive User Name

User name for the Hive connection.

Select Hive Execution Mode

Select the Hive execution mode. Default is Remote.

Hadoop Impersonation User Name

User name of the impersonation user.

Note: The Data Integration Service uses this user name to execute mappings on a secure cluster.

Hadoop Blaze Working Directory

Path to the Blaze working directory on the HDInsight cluster.

For example,

```
/blaze/workdir
```

Hadoop Blaze Service User Name

User name for the Blaze engine service on the HDInsight cluster.

Hadoop Spark HDFS Staging Directory

Path to the Spark HDFS staging directory on the HDInsight cluster.

For example,

```
/tmp/sparkdir
```

Hadoop Spark Event Log Directory

Path to the Spark HDFS event log directory on the HDInsight cluster.

Hadoop Spark Execution Parameters

Optional. Run-time parameters that override Spark parameters on the cluster. You can enter a list separated by the characters "&:" (ampersand colon).

For example, when you want to set run-time parameters for the spark.driver.memory, spark.executor.memory, and spark.executor.cores properties, enter a string like:

```
Spark.driver.memory=1G&:spark.executor.memory=4G&:spark.executor.cores=2
```

9. Supply information in the **Infrastructure Settings** panel, and then click **OK**.

   Use this tab to set up cluster resources for the Big Data Management implementation.

   Storage account

   Storage resource that the virtual machines that run the Big Data Management implementation will use for data storage.

   Select from the existing storage accounts.

   Virtual network

   Virtual network for the Big Data Management implementation to belong to. Select the same network as the one that you used to create the HDInsight cluster.

   Subnets

   The subnet that the virtual network contains.

   Choose from among the subnets that are available in the virtual network.

10. Verify the choices in the **Summary** panel, and then click **OK**.
11. Read the terms of use in the **Buy** panel, and then click **Create**.

    When you click **Create**, Azure deploys Big Data Management and creates resources in the environment that you configured.

Logs and Results Files

The installer creates the following logs and results files:

- Informatica installation logs. Location: `/home/<operating system user name>/Informatica/10.1.1`

  - `Informatica_10.1.1_InstallLog.log`

  - `Informatica_10.1.1_Services_<time stamp>.log`

- Big Data Management solution deployment log. Location: `/home/<operating system user name>/Oneclicksoluion_results.log`
- The Big Data Management installer runs from a Debian package. The log for this operation is in the following location on each node: `/home/<operating system user name>/debianresults`

Post-Implementation: Import Files for Data Decryption

If you configured the SSH public key user as a user of the Big Data Management instance, perform the following task when the Big Data Management installer completes.

The node that hosts the Data Integration Service requires the following files from the Hadoop cluster to decrypt data:

- key_decryption_cert.prv

- decrypt.sh

Perform the following steps to import the files:

1. Locate these files on the cluster head node.
2. Copy the files from the Hadoop cluster to the node that runs the Data Integration Service. The directory structure must be the same as that of the Hadoop cluster.
3. Verify that permissions on the directory are 775.

Next Steps

To see how to use Big Data Management, read Big Data Management documentation.

Each of these guides is available in the Big Data Management documentation set on the Informatica Network at http://network.informatica.com.

Informatica Application Service Guide

Describes the Model Repository Service, Data Integration Service, and other application services that Big Data Management uses.

Big Data Management User Guide

Describes how to use Informatica Developer and Informatica Administrator to manage connections between the Informatica domain and the cluster, and how to create mappings in the Developer tool.

Informatica Developer User Guide

Contains full details about how to use the Developer tool to create and run mappings and workflows.

Informatica Developer Mapping Guide

Contains full details about how to develop mappings in the Developer tool.

Author

Mark Pritchard